

ORAL AND SPOTLIGHT PRESENTATIONS

Oral Session 1: Generative modelling: Image Reconstruction and Synthesis Monday, October 7, 2024, 11:30 to 12:30 Orion Tent (Main Hall) Session Chairs: Bernhard Kainz, Imperial College London, UK and FAU Erlangen-Nürnberg, Germany Can Zhao, Nvidia, United States	Oral Session 2: Surgical Scene Understanding Monday, October 7, 2024, 11:30 to 12:30 Crystal Room (Dual Track Hall) Session Chairs: Sophia Bano, University College London, UK Mathias Unberath, Johns Hopkins University, USA
Myocardial Scar Enhancement in LGE Cardiac MRI using Localized Diffusion Speaker: Marta Hasny, Technical University of Munich, Germany Center-to-Edge Denoising Diffusion Probabilistic Models with Cross-domain Attention for Undersampled MRI Reconstruction Speaker: Shuo Li, Case Western Reserve University, USA Masked Residual Diffusion Probabilistic Model with Regional Asymmetry Prior for Generating Perfusion Maps from Multi-phase CTA Speaker: Yuxin Cai, Huazhong University of Science and Technology, China Diffusion as Sound Propagation: Physics-inspired Model for Ultrasound Image Generation Speaker: Marina Domínguez, MCML, Germany	Depth-Driven Geometric Prompt Learning for Laparoscopic Liver Landmark Detection Speaker: Jialun Pei, The Chinese University of Hong Kong, Hong Kong SAR Enhanced Scale-aware Depth Estimation for Monocular Endoscopic Scenes with Geometric Modeling Speaker: Ruofeng Wei, The Chinese University of Hong Kong, Hong Kong SAR Structure-preserving Image Translation for Depth Estimation in Colonoscopy Speaker: Shuxian Wang, UNC Chapel Hill, USA Transferring Relative Monocular Depth to Surgical Vision with Temporal Consistency Speaker: Charlie Budd, King's College London, UK

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Oral & Spotlight Session 3: Machine Learning Strategies for MICCAI Monday, October 7, 2024, 13:30 to 15:00 Orion Tent (Main Hall) Session Chairs: Olivier Colliot, CNRS, France Sahar Selim Soussa, Nile University, Egypt	Oral & Spotlight Session 4: Health Equity: Care for All Monday, October 7, 2024, 13:30 to 15:00 Crystal Room (Dual Track Hall) Session Chairs: Xiaoxiao Li, University of British Columbia, Canada Saad Nadeem, Memorial Sloan Kettering Cancer Center, USA
<u>Oral Presentations:</u> Diffusion-based Domain Adaptation for Medical Image Segmentation using Stochastic Step Alignment Speaker: Wen Ji, Hong Kong University of Science and Technology, Hong Kong SAR CoReEcho: Continuous Representation Learning for 2D+time Echocardiography Analysis Speaker: Fadillah Maani, Mohamed Bin Zayed University of Artificial Intelligence, United Arab Emirates Few Slices Suffice: Multi-Faceted Consistency Learning with Active Cross-Annotation for Barely-supervised 3D Medical Image Segmentation Speaker: Zhe Xu and Xinyao Wu, The Chinese University of Hong Kong, Hong Kong SAR Self-Supervised k-Space Regularization for Motion-Resolved Abdominal MRI Using Neural Implicit k-Space Representations Speaker: Veronika Spieker, Helmholtz Munich / Technical University of Munich, Germany A Clinical-oriented Lightweight Network for High-resolution Medical Image Enhancement Speaker: Osmar R. Zaiane, Alberta Machine Intelligence Institute, University of Alberta, Canada	<u>Oral Presentations:</u> S-SYNTH: Knowledge-Based, Synthetic Generation of Skin Images Speaker: Niloufar Saharkhiz, FDA, USA PASSION for Dermatology: Bridging the Diversity Gap with Pigmented Skin Images from Sub-Saharan Africa Speaker: Philippe Gottfrois, University Basel/ University Hospital Basel, Switzerland Towards Rapid Mycetoma Species Diagnosis: A Deep Learning Approach for Stain-Invariant Classification on H&E Images from Senegal Speaker: Kpêchéhoué Merveille Santi ZINSOU, University of Gaston Berger, Senegal FD-SOS: Vision-Language Open-Set Detectors for Bone Fenestration and Dehiscence Detection from Intraoral Images Speaker: Marawan Elbatel, The Hong Kong University of Science and Technology, Hong Kong SAR EchoMEN: Combating Data Imbalance in Ejection Fraction Regression via Multi-Expert Network Speaker: Song Lai, City University of Hong Kong, Hong Kong SAR
<u>Spotlight Presentations:</u> Advancing Text-Driven Chest X-Ray Generation with Policy-Based Reinforcement Learning Speaker: Woojung Han, Yonsei University, South Korea Few-shot Adaptation of Medical Vision-Language Models Speaker: Julio Silva-Rodríguez, École de technologie supérieure (ÉTS), Canada Physics informed neural networks for estimation of tissue properties from multi-echo configuration state MRI Speaker: Samuel Adams-Tew, University of Utah, USA	<u>Spotlight Presentations:</u> RadiomicsFill-Mammo: Synthetic Mammogram Mass Manipulation with Radiomics Features Speaker: Inye Na, Sungkyunkwan University, South Korea FairDiff: Fair Segmentation with Point-Image Diffusion Speaker: Wenyi Li, Institute for AI Industry Research (AIR), Tsinghua University, China SlicerTMS: Real-Time Visualization of Transcranial Magnetic Stimulation for Mental Health Treatment Speaker: Loraine Franke, University of Massachusetts Boston, USA

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Oral Session 5: Computer Aided Diagnosis Monday, October 7, 2024, 16:30 to 18:00 Orion Tent (Main Hall) Session Chairs: Ulas Bagci, Northwestern University, USA Lei Li, University of Southampton, United Kingdom	Oral Session 6: Clinical Translation: Functional Imaging and Oncology Monday, October 7, 2024, 16:30 to 18:00 Crystal Room (Dual Track Hall) Session Chairs: Jana Hutter, King's College London, UK Wenjia Bai, Imperial College London, UK
Aligning Medical Images with General Knowledge from Large Language Models Speaker: Yi Lin, The Hong Kong University of Science and Technology, Hong Kong SAR, China Cardiovascular Disease Detection from Multi-View Chest X-rays with BI-Mamba Speaker: Pingkun Yan, Rensselaer Polytechnic Institute, USA Deep Learning for Cancer Prognosis Prediction Using Portrait Photos by StyleGAN Embedding Speaker: Amr Hagag, University Hospital Erlangen, Germany Topological GCN for Improving Detection of Hip Landmarks from B-Mode Ultrasound Images Speaker: Jun Shi, Shanghai University, China TAPoseNet: Teeth Alignment based on Pose estimation via multi-scale Graph Convolutional Network Speaker: Qingxin Deng, Shenzhen University, China 3D Spine Shape Estimation from Single 2D DXA Speaker: Emmanuelle Bourigault, University of Oxford, UK	Biophysics-based data assimilation of longitudinal tau and amyloid-β PET scans Speaker: George Biros, The University of Texas at Austin, USA MetaAD: Metabolism-Aware Anomaly Detection for Parkinson's Disease in 3D 18F-FDG PET Speaker: Zhenrong Shen, Shanghai Jiao Tong University, China Genomics-guided Representation Learning for Pathologic Pan-cancer Tumor Microenvironment Subtype Prediction Speaker: Fangliangzi Meng, Tongji University, China Ordinal Learning: Longitudinal Attention Alignment Model for Predicting Time to Future Breast Cancer Events from Mammograms Speaker: Xin Wang, The Netherlands Cancer Institute, The Netherlands MMFusion: Multi-modality Diffusion Model for Lymph Node Metastasis Diagnosis in Esophageal Cancer Speaker: Chengyu Wu, Shandong University, China Is this hard for you? Personalized human difficulty estimation for skin lesion diagnosis Speaker: Peter Johannes Tejlgaard Kampen, Technical University of Denmark, Denmark

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Oral Session 7: Health Equity: Low Resource Settings Tuesday, October 8, 2024, 8:30 to 9:30 Orion Tent (Main Hall) Session Chairs: Chen (Cherise) Chen, University of Sheffield, UK Prateek Prasanna, Stony Brook University, USA	Oral Session 8: Image Registration Tuesday, October 8, 2024, 8:30 to 9:30 Crystal Room (Dual Track Hall) Session Chairs: Jelmer Wolterink, University of Twente, Netherlands Fatemeh Zabihollahy, University of Toronto, Canada
TinyU-Net: Lighter yet Better U-Net with Cascaded Multi-Receptive Fields Speaker: Junren Chen, Sichuan University, China UnWave-Net: Unrolled Wavelet Network for Compton Tomography Image Reconstruction Speaker: Ishak Ayad, ENSEA & CY Cergy Paris University, France Rethinking Histology Slide Digitization Workflows for Low-Resource Settings Speaker: Talat Zehra, Jinnah Sindh medical university, Pakistan SpeChrOmics: A Biomarker Characterization Framework for Medical Hyperspectral Imaging Speaker: Ajibola S. Oladokun, University of Cape Town, South Africa	Large-Scale 3D Infant Face Model Speaker: Till Schnabel, ETH Zürich, Switzerland NODER: Image Sequence Regression Based on Neural Ordinary Differential Equations Speaker: Yi Hong, Shanghai Jiao Tong University, China Aligning and Restoring Imperfect ssEM images for Continuity Reconstruction Speaker: Haiyang Yan, Institute of Automation, Chinese Academy of Sciences, China IM-MoCo: Self-supervised MRI Motion Correction using Motion-Guided Implicit Neural Representations Speaker: Ziad Al-Haj Hemidi, Universität zu Lübeck, Germany

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Oral & Spotlight Session 9: Image Segmentation Tuesday, October 8, 2024, 13:30 to 15:00 Orion Tent (Main Hall) Session Chairs: Masahiro Oda, Nagoya University, Japan Chen Qin, Imperial College London, UK	Oral & Spotlight Session 10: Foundation Models and Multimodal Data for MICCAI Tuesday, October 8, 2024, 13:30 to 15:00 Crystal Room (Dual Track Hall) Session Chairs: Davood Karimi, Harvard University, USA Nadieh Khalili, Radboud UMC, Netherlands
<u>Oral Presentations:</u> DPMNet: Dual-Path MLP-based Network for Aneurysm Image Segmentation Speaker: Xue Zhao, China University of Petroleum (East China), China HRDecoder: High-Resolution Decoder Network for Fundus Image Lesion Segmentation Speaker: Ziyuan Ding, Central South University, China PEMMA: Parameter-Efficient Multi-Modal Adaptation for Medical Image Segmentation Speaker: Nada Saadi, Mohamed Bin Zayed University of Artificial Intelligence, United Arab Emirates CT-based brain ventricle segmentation via diffusion Schrödinger Bridge without target domain ground truths Speaker: Yiming Xiao, Concordia University, Canada IterMask2: Iterative Unsupervised Anomaly Segmentation via Spatial and Frequency Masking for Brain Lesions in MRI Speaker: Ziyun Liang, University of Oxford, UK	<u>Oral Presentations:</u> Whole Heart 3D+T Representation Learning Through Sparse 2D Cardiac MR Images Speaker: Yundi Zhang, AI in Medicine and Healthcare/Technical University of Munich, Germany ProstNFound: Integrating Foundation Models with Ultrasound Domain Knowledge and Clinical Context for Robust Prostate Cancer Detection Speaker: Paul Wilson, Queen's University, Canada DB-SAM: Delving into High Quality Universal Medical Image Segmentation Speaker: Chao Qin, Mohamed Bin Zayed University of Artificial Intelligence, United Arab Emirates Learning to Segment Multiple Organs from Multimodal Partially Labeled Datasets Speaker: Yefeng Zheng, Tencent, China M4oE: A Foundation Model for Medical Multimodal Image Segmentation with Mixture of Experts Speaker: Yiqing Shen, Johns Hopkins University, USA
<u>Spotlight Presentations:</u> FRCNet: Frequency and Region Consistency for Semi-supervised Medical Image Segmentation Speaker: Huazhu Fu, Institute of High Performance Computing (IHPC), Agency for Science, Technology and Research (A*STAR), Singapore SegMamba: Long-range Sequential Modeling Mamba For 3D Medical Image Segmentation Speaker: Zhaohu Xing, The Hong Kong University of Science and Technology (Guangzhou), China Swin SMT: Global Sequential Modeling for Enhancing 3D Medical Image Segmentation Speaker: Maciej Chrabaszcz, Warsaw University of Technology, Poland	<u>Spotlight Presentations:</u> Black-Box Adaptation for Medical Image Segmentation Speaker: Jay N. Paranjape, Johns Hopkins University, USA DRIM: Learning Disentangled Representations from Incomplete Multimodal Healthcare Data Speaker: Lucas Robinet, Oncopole Claudius Regaud, France An approach to building foundation models for brain image analysis Speaker: Davood Karimi, Harvard Medical School, USA

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Oral Session 11: Transparency, Fairness and Uncertainty Tuesday, October 8, 2024, 16:30 to 18:00 Orion Tent (Main Hall) Session Chairs: Tanveer Syeda-Mahmood, IBM Research, USA Lequan Yu, University of Hong Kong, Hong Kong SAR	Oral Session 12: Surgical Data Science Tuesday, October 8, 2024, 16:30 to 18:00 Crystal Room (Dual Track Hall) Session Chairs: Ruogu Fang, University of Florida, United States Mobarakol Islam, University College London, UK
A Large-scale Multi Domain Leukemia Dataset for the White Blood Cells Detection with Morphological Attributes for Explainability Speaker: Abdul Rehman, Information Technology University, Pakistan Follow the Radiologist: Clinically Relevant Multi-View Cues for Breast Cancer Detection from Mammograms Speaker: Kshitiz Jain, Indian Institute of Technology Delhi, India Learning a Clinically-Relevant Concept Bottleneck for Lesion Detection in Breast Ultrasound Speaker: Arianna Bunnell, University of Hawaii Cancer Center, USA BiasPruner: Debiased Continual Learning for Medical Image Classification Speaker: Nourhan Bayasi, University of British Columbia, Canada Are We Ready for Out-of-Distribution Detection in Digital Pathology? Speaker: Ji-Hun Oh, UIUC, USA Geometric Transformation Uncertainty for Improving 3D Fetal Brain Pose Prediction from Freehand 2D Ultrasound Videos Speaker: Jayroop Ramesh, University of Oxford, UK	PitVQA: Image-grounded Text Embedding LLM for Visual Question Answering in Pituitary Surgery Speaker: Runlong He, University College London, UK ORacle: Large Vision-Language Models for Knowledge-Guided Holistic OR Domain Modeling Speaker: Ege Özsoy, TUM, Germany Deep intra-operative illumination calibration of hyperspectral cameras Speaker: Alexander Baumann, Siemens AG/German Cancer Research Center/Heidelberg University, Germany SALI: Short-term Alignment and Long-term Interaction Network for Colonoscopy Video Polyp Segmentation Speaker: Qiang Hu, Huazhong University of Science and Technology, China VideoCutMix: Temporal Segmentation of Surgical Videos in Scarce Data Scenarios Speaker: Rohan Raju Dhanakshirur, Indian Institute of Technology, Delhi, India TeleOR: Real-time Telemedicine System for Full-Scene Operating Room Speaker: Yixuan Wu, Zhejiang University, China

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Oral Session 13: Neuroimaging Wednesday, October 9, 2024, 8:30 to 9:30 Orion Tent (Main Hall) Session Chairs: Samuel Kadoury, Polytechnique Montréal, Canada Anees Kazi, Harvard Medical School, USA	Oral Session 14: Computational Pathology Wednesday, October 9, 2024, 8:30 to 9:30 Crystal Room (Dual Track Hall) Session Chairs: Anne Martel, Sunnybrook Research Institute, Canada Sang Hyun Park, Daegu Gyeongbuk Institute of Science and Technology, Korea
Enhancing Spatiotemporal Disease Progression Models via Latent Diffusion and Prior Knowledge Speaker: Lemuel Puglisi, University of Catania, Italy Longitudinally Consistent Individualized Prediction of Infant Cortical Morphological Development Speaker: Xinrui Yuan, University of North Carolina at Chapel Hill, USA BrainWaveNet: Wavelet-based Transformer for Autism Spectrum Disorder Diagnosis Speaker: Ah-Yeong Jeong, Korea University, South Korea TractOracle: towards an anatomically-informed reward function for RL-based tractography Speaker: Antoine Théberge, Université de Sherbrooke, Canada	WsiCaption: Multiple Instance Generation of Pathology Reports for Gigapixel Whole-Slide Images Speaker: Pingyi Chen, Westlake University, China DSCENet: Dynamic Screening and Clinical-Enhanced Multimodal Fusion for MPNs Subtype Classification Speaker: Yuan Zhang, Southeast University, China Joint multi-task learning improves weakly-supervised biomarker prediction in computational pathology Speaker: Omar S. M. El Nahhas, EKFZ for Digital Health, Germany Enhancing Gene Expression Prediction from Histology Images with Spatial Transcriptomics Completion Speaker: Gabriel Mejia and Daniela Ruiz, Center for Research and Formation in Artificial Intelligence (CinfonIA), Universidad de los Andes, Colombia

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Oral & Spotlight Session 15: Computer Assisted Interventions Wednesday, October 9, 2024, 13:30 to 15:00 Orion Tent (Main Hall) Session Chairs: Tina Kapur, Brigham and Women's Hospital, USA Yoshito Otake, Nara Institute of Science and Technology, Japan	Oral & Spotlight Session 16: Clinical Translation: Neuro, Spine and Ultrasound Wednesday, October 9, 2024, 13:30 to 15:00 Crystal Room (Dual Track Hall) Session Chairs: Syed Zulqarnain Gilani, Edith Cowan University, Australia Archana Venkataraman, Johns Hopkins University, USA
<u>Oral Presentations:</u> Topological SLAM in colonoscopies leveraging deep features and topological priors Speaker: Javier Morlana, University of Zaragoza, Spain EndoSelf: Self-Supervised Monocular 3D Scene Reconstruction of Deformable Tissues with Neural Radiance Fields on Endoscopic Videos Speaker: Wenda Li, Nagoya University, Japan LighTDiff: Surgical Endoscopic Image Low-Light Enhancement with T-Diffusion Speaker: Tong Chen, The University of Sydney, Australia Keypoint Matching for Instrument-Free 3D Registration in Video-based Surgical Navigation Speaker: Tânia Baptista, Universidade de Coimbra, Portugal Transforming Surgical Interventions with Embodied Intelligence for Ultrasound Robotics Speaker: Huan Xu, CAIR, HKISI-CAS, Hong Kong SAR	<u>Oral Presentations:</u> RoCoSDF: Row-Column Scanned Neural Signed Distance Fields for Freehand 3D Ultrasound Imaging Shape Reconstruction Speaker: Hongbo Chen, ShanghaiTech University, China Hybrid-Structure-Oriented Transformer for Arm Musculoskeletal Ultrasound Segmentation Speaker: Lingyu Chen, Nanjing University of Aeronautics and Astronautics, China Quantitative Assessment of Thyroid Nodules through Ultrasound Imaging Analysis Speaker: Young-Min Kim, KAIST, South Korea SimBrainNet: Evaluating Brain Network Similarity for Attention Disorders Speaker: Debashis Das Chakladar, Luleå University of Technology, Sweden Knowledge-Guided Prompt Learning for Lifespan Brain MR Image Segmentation Speaker: Lin Teng, ShanghaiTech University, China
<u>Spotlight Presentations:</u> Vision-Based Neurosurgical Guidance: Unsupervised Localization and Camera-Pose Prediction Speaker: Gary Sarwin, ETH Zurich, Switzerland Two Projections Suffice for Cerebral Vascular Reconstruction Speaker: Alexandre Cafaro, Université Paris-Saclay, France DCrownFormer: Morphology-aware Point-to-Mesh Generation Transformer for Dental Crown Prosthesis from 3D Scan Data of Antagonist and Preparation Teeth Speaker: Su Yang, Seoul National University, South Korea	<u>Spotlight Presentations:</u> PRISM: A Promptable and Robust Interactive Segmentation Model with Visual Prompts Speaker: Hao Li, Vanderbilt University, USA A Patient-Specific Framework for Autonomous Spinal Fixation via a Steerable Drilling Robot Speaker: Susheela Sharma, The University of Texas at Austin, USA Automated Spinal MRI Labelling from Reports Using a Large Language Model Speaker: Robin Y. Park, University of Oxford, UK